



Name: _____ Cohort/Batch: _____ Date: _____ Score: _____

KOA PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 Q&A on async middleware, ctx object, routing, and error handling

TOTAL: 10 QUESTIONS

Q1 What is Koa and how does it improve on Express's middleware model?

Q6 How does centralized error handling work in Koa?

Q2 How does Koa's cascading middleware work in practice?

Q7 How do you parse request bodies in Koa?

Q3 What is the Koa ctx (context) object?

Q8 How do you manage sessions in Koa?

Q4 What is the difference between ctx.body and ctx.response.body?

Q9 How do you serve static files in Koa?

Q5 How do you handle routing in Koa with koa-router?

Q10 How do you test a Koa application?



ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 Koa is built by the Express team

Mitigation

Koa is built by the Express team using `async/await` instead of callbacks. Its cascade middleware uses `async` functions and `await next()`, giving you clean before/after control without nesting. Koa has no bundled routing or body parsing use separate packages.

A6 Add an error-catching middleware as the first

Observability

Add an error-catching middleware as the first in the stack: `try { await next() } catch (err) { ctx.status = err.status || 500; ctx.body = {error: err.message}; ctx.app.emit('error', err, ctx) }`. It catches errors from all downstream middleware.

A2 Middleware runs sequentially until `next()` is awaited,

Primitives

Middleware runs sequentially until `next()` is awaited, then in reverse after. Code before `await next()` runs on the way in; code after runs on the way out. This makes it trivial to measure duration: `record time before next(), log delta after.`

A7 koa-bodyparser adds `ctx.request.body` for JSON and

Automation

`koa-bodyparser` adds `ctx.request.body` for JSON and URL-encoded bodies. Register it before routes: `app.use(bodyParser())`. For multipart file uploads use `@koa/multer` or busboy-based middleware. `koa-body` combines both.

A3 `ctx` bundles the Node.js request and response:

Hardening

`ctx` bundles the Node.js request and response: `ctx.request` (Koa's enhanced request), `ctx.response`, `ctx.params` (router), `ctx.state` (app-level data store), and shortcuts like `ctx.body`, `ctx.status`, `ctx.get()`, `ctx.set()`. Middleware reads and writes via `ctx`.

A8 koa-session adds `ctx.session` to store per-user data

Governance

`koa-session` adds `ctx.session` to store per-user data in a signed cookie. For large payloads configure an external store (Redis). `app.keys = ['secret']` must be set for cookie signing. Session data is loaded on request and saved on response.

A4 They are the same underlying property

Gotchas

They are the same underlying property. `ctx.body` is an alias for `ctx.response.body`. Koa delegates many properties from `ctx` directly to `ctx.request` and `ctx.response` as shortcuts. Setting `ctx.body` to an object automatically sends JSON with the correct Content-Type.

A9 koa-static(`root`, `options`) creates middleware that checks if

Performance

`koa-static(root, options)` creates middleware that checks if the URL maps to a file in `root`, reads it, and sets the correct Content-Type. Register it early so static file requests bypass the router entirely. Set `maxage` for browser caching.

A5 `const router = new Router(); router.get('/users/:id', async`

Iteration

`const router = new Router(); router.get('/users/:id', async ctx => { ctx.body = await getUser(ctx.params.id) });` Use `router.routes()` and `router.allowedMethods()` as middleware on the Koa app. Groups can share prefixes with `router.prefix('/api')`.

A10 Use supertest: `const server = app.callback(); await`

Optimization

Use supertest: `const server = app.callback(); await request(server).get('/users').expect(200)`. `app.callback()` returns a plain `http.RequestListener` without starting a server, allowing tests to use ephemeral ports managed by supertest.